SoK: Transfer Boundaries of LLM and Agent Side-Channel Defenses

***Abstract*—Side-channel defenses for LLM serving and agent systems are evaluated at one operating point and reused at another. This SoK systematizes the evidence required to carry a defense result across deployment boundaries. Our unit is the evaluated result record: one outcome bound to its target, receiver, intervention, attacker fit, workload and cost object. Across 82 study families and 39 records, eight of 15 same-family pairs concern different primitives, targets or costs; only 17 of 133 related pairs support a source-qualified descriptive comparison. Record-local joins identify one configuration-linked inference–utility–cost package, supplied by two CachePrune records. A six-axis holdout map distinguishes evidence for input, model, topology, hardware, network and temporal changes. Four controlled studies support the synthesis with costed examples of emitter, execution-state, calibration and observation changes. Their results expose distinct re-evaluation needs: timing can recover signal missed by length-only attacks, while 100 training-customer resamples put local GLM return accuracy at 50.0–67.9%, preventing a model-source interpretation of an unstable calibration contrast. We integrate the literature audit and these scoped measurements into a deployment decision table: identify the changed boundary, retain applicable evidence, and measure the missing binding before reusing a defense claim.**

1. Introduction

Encryption conceals message content, yet LLM engines and agents expose packet sizes, cache timing, visited domains and tool sequences [1], [2], [3], [4], [5], [6]. Defenses follow: padding, batching, cache isolation, decoy queries and action humanization. Each is evaluated at one receiver, execution boundary and workload, yet its headline number is cited as a property of the mechanism. A padding result measured on an upstream link says nothing about a frontend that re-emits decoded text; decoys whose read responses are discarded say nothing about cover that executes writes; and the overhead of a semantic cache does not price prefix isolation reported in the same paper.

We therefore systematize defense evidence at the level of the *evaluated result record* rather than the paper. A record binds one outcome to its target, receiver projection, intervention boundary, attacker fit, workload and cost object (§3). Three instruments operate on these records: a map of 82 study families by signal origin, receiving view and threat polarity; a nine-item audit of what each mitigation family binds; and a six-axis matrix of reported holdouts. Together they answer one question: *which changes invalidate a measured defense result?*

A SoK of defense-evidence transfer. Our central contribution is a record-level synthesis of when defense evidence can be reused. Controlled studies support this synthesis by measuring selected deployment changes. The family map locates the changes, the record audit establishes which outcomes can be joined, and the studies attach resource and behavioral consequences to the resulting decisions. This yields three connected contributions:

- **Identify the evidence that transfers (§§2–6).** The complete 39-record audit finds object substitutions in 8 of 15 same-family pairs; 17 of 133 related pairs support a source-qualified descriptive comparison. Configuration-linked joins identify CachePrune's single inference–utility–cost package through two records. The 82-family, six-axis holdout map distinguishes reported input separation from evidence for a changed generator or environment.
- **Measure the consequences of selected boundary changes (§8).** Four controlled studies connect emitter placement, stateful execution, calibration and attacker observation to scoped inference, utility and resource costs. Emitter/state studies reproduce mechanisms and measure their operating consequences; calibration and timing studies examine local training variability, residual operation signatures and live execution conditions.
- **Turn the synthesis into deployment decisions (Table 11).** Each changed boundary has a minimum re-evaluation requirement. For calibration, the 100-resample study establishes the local baseline needed before attributing a contrast to its source model. For execution, a fresh 100-episode planner study measures prefix-cover success under response deadlines. These measurements support decisions at their own operating points.

WF practice establishes adaptive attacks, distribution shifts and online-cost evaluation [7], [8]. We extend this evaluation discipline to evidence reuse across LLM receivers, tool-state boundaries and planner behavior, using configuration-linked records and a deployment decision table. Table 1 connects each question to its literature synthesis and supporting measurement.

TABLE 1. ONE SOK QUESTION—WHICH DEFENSE EVIDENCE SURVIVES A DEPLOYMENT CHANGE? LITERATURE SYNTHESIS SUPPLIES THE ANSWER STRUCTURE; CONTROLLED STUDIES MEASURE SELECTED CONSEQUENCES.

Question	Systematized finding	Evidence and supporting study
RQ1: targets and observables	Signal generation clusters at serving (32 families) and orchestration (19), while receiving views cluster at application, hardware and transport. Receiver and threat polarity determine which protection can be reused.	Complete 58-core-family map (§4); final-emitter measurement in Exp 1.
RQ2: comparable outcomes	Eight of 15 same-family pairs concern different primitives, targets or costs; only 17 of 133 related pairs support descriptive comparison. Shared mechanism names do not bind execution or state effects.	Complete 39-record pair audit (§5); state/link measurements in Exp 2.
RQ3: transfer axes	Reported task/input holdouts (13 families) and model holdouts (three) answer different reuse questions. Hardware/time entries are mostly unresolved; calibration contrasts also require a local variability baseline.	82-family six-axis matrix (§6, Appendix D); source/local-resampling comparison in Exp 3.
RQ4: joint evidence	CachePrune supplies one configuration-linked inference–utility–cost package through two records. Evidence joins require a common operating point and cost object.	351 record–item cells and 20-family audit (§5); timing, RPC and planner measurements in Exps 3–4; deployment decisions in Table 11.

Scope of claims. The real-agent target is a requested operation category in one retail environment with a known 53-customer cohort. Measurements establish fitted-suite inference, logical state conformance and scoped resource costs; a separate 100-episode live study measures prefix-cover utility under injected deadlines. Deployment-wide security and full joint utility require the validation specified in §9.3.

2. Scope and Systematization Method

2.1. Research questions, eligibility and threat polarity

We organize the systematization around four questions: which hidden targets, threat polarities and auxiliary observables appear across LLM and agent systems (RQ1); how signal generation origin relates to receiver observation permissions (RQ2); which holdouts establish portability across tasks, configurations and hardware (RQ3); and which evidence binds workflow protection, logical state preservation and resource cost, distinguishing these from full end-to-end planner utility (RQ4).

Core inclusion requires five conjunctive criteria: (C1) LLM or agent execution, (C2) a hidden target property, (C3) an execution-dependent auxiliary observable rather than ordinary output semantics alone, (C4) specified observer permissions, and (C5) an empirical inference evaluation. Mixed-feature papers qualify only if they isolate an auxiliary-only C3 component; thus coding fingerprints, WildAuth, ZeroPrint, endpoint-content fingerprints and StepFinder remain adjacent boundary comparators [9], [10], [11], [12], [13]. Table 2 states the observable eligibility rules.

We use four narrative groupings of objectives and observer roles, not mutually exclusive classes. An unauthorized identity inference can be both attribution and a confidentiality concern. Each record separately specifies the inference target, observer, authorization relation and protected interest: **Type I** concerns confidentiality/privacy; **Type II** concerns model, harness or identity attribution; **Type III** concerns evasion of a platform's human–agent detector; **Type IV** concerns authorized operator diagnostics. Types can overlap. WRA protects research intent (I); UI Traces attributes agents (II); AHB evades a defensive detector (III); ClawGuard diagnoses loops (IV) [6], [5], [14], [15]. These roles are mapped per core family, alongside generating and receiving layers, in Table 6.

TABLE 2. OBSERVABLE-BASED ELIGIBILITY. A HIDDEN TARGET ALONE DOES NOT MAKE A SEMANTIC ATTACK A SIDE CHANNEL.

Observed input	Decision and required distinction
Length, timing, cache response	Core when measured execution dependence supports hidden-target inference.
Own-site actions	Core attribution with site permission; no automatic access to other sites or private tool logs.
Reply states the secret	Direct disclosure; adjacent, not auxiliary-channel evidence.
Reply style/content identifies a user	Adjacent unless a separately evaluated auxiliary mechanism is established. WildAuth stays a mixed-view comparator.
Output-filter membership oracle	Can be core: the filter's execution, not ordinary semantic output inference, mediates the signal.
Privileged loop/fault logs	Separate diagnostic component when C1–C5 hold; does not increase confidentiality evidence.

2.2. Corpus construction and reconciliation

Forty seeds, venue-targeted proceedings screening and one-hop citation snowballing produced an initial map of 60 families (43 core, 17 adjacent). A supplementary database and bulk keyword search returned 615 occurrences and 565 unique records; auditing its 204 exclusions rerouted four records, and the search added 15 core and seven adjacent families. The final collection has 58 core and 24 adjacent families. The completed search frames, closed on October 21, 2026, combine Semantic Scholar and native arXiv queries with venue-targeted screening of S&P, USENIX Security, NDSS, CCS and PMLR. C1–C5 screening and version deduplication operate on records before family reconciliation; the supplement retains queries, dispositions and excluded incomplete search frames.

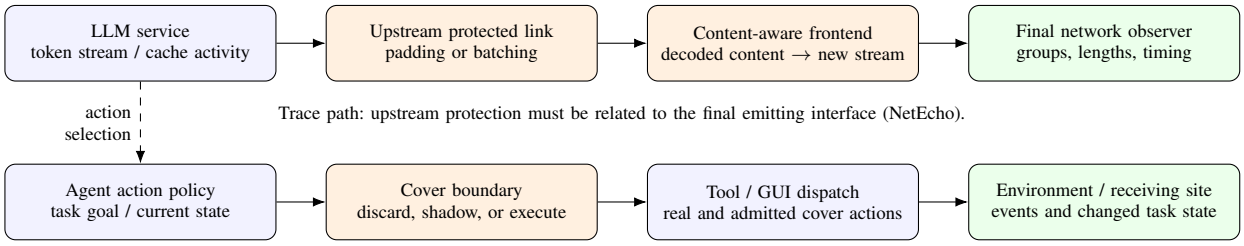


Figure 1. Observation and execution boundaries. The controlled studies support the SoK by measuring selected emitter, execution, calibration and observation changes; the record audit determines the scope of evidence reuse.

2.3. Source-bound result records

Each result record binds its target, observation boundary, operating point and source section/table/figure/paragraph locator. The original records and family audit entries were author-checked against these sources. The record-local extension reuses these bindings and documents targeted source-follow-up changes. Appendix B indexes all 39 records with their measurement modes; Appendix A describes the supporting artifact.

SoK Takeaway 1 (Map evidence to the receiving deployment). A family-level label does not identify which signal, observer permission or protected objective a result supports. The complete core map separates signal origin from receiving access; the result ledger then binds each comparison to a source version and operating point.

3. Result Records and Comparability Rules

The atomic unit of our systematization is one *evaluated result record*, rather than a whole paper or a single headline score. A result record binds a source version and locator to seven structured fields: Source, Target (and threat polarity Types I–IV), Receiver projection, Intervention boundary, Attacker training/calibration (D_0: clean-condition fit; D_2: protected-condition fit), Workload/holdout, and Outcomes (inference metric, task/component utility, and cost object; Table 3). A study family contributes multiple records whenever it evaluates distinct execution modes (e.g., stored capture vs. live loopback in Dangol [16]), distinct mechanisms (e.g., K-prefix KV simulation vs. GPTCache semantic de-identification in Early Bird [3]), or separately scoped performance benchmarks (e.g., SafeKV [17] and OTRO [18]).

Indexing schema and comparison contracts. The mnemonic SCOPE (Secret/target, Channel, Observer, Pipeline, Exploitation goal) names the target-and-channel columns of our index, paired with evaluation provenance. Generating layers span hardware (P1), serving (P2), transport (P3), application/API (P4), orchestration (P5), tools/MCP (P6), environment (P7) and multi-agent communication (P8). We apply three comparison contracts (Table 4): a comparison is *compatible* when required fields align; *incompatible* when a positive mechanism or boundary conflict is established (such as substituting token-payload ratios for wire bytes or offline timestamp edits for live action delays); and *unresolved* when a required binding is unreported (U), unavailable (A) or unverified (?). Table 5 illustrates a complete record-to-judgment lineage.

From contracts to an audit. Section 5 operationalizes these contracts as nine reuse items with a shared evidence-status alphabet (Table 8) and applies them to every mitiga-

TABLE 3. A SINGLE RESULT RECORD. ONLY FIELDS RELEVANT TO THE REQUESTED COMPARISON MUST ALIGN; UNAVAILABLE REQUIRED FIELDS REMAIN UNRESOLVED.

Record field	Required meaning
Source	Family/result ID, version, source locator, reading scope and hash.
Target	Hidden property, inference goal and protection relation: confidentiality, attribution, human/agent distinction or authorized diagnosis.
Receiver	Actual permission and released representation, including session/document linkage and any auxiliary inputs.
Intervention	Where protection ends; whether edits affect stored traces, transmission or executed actions.
Attacker	Fit/calibration condition, test population, training access and adaptation budget.
Workload	Task, system, topology/hardware and grouping/holdout.
Outcomes	Inference metric / denominator; objective or proxy utility; applicable cost object; uncertainty and field states.

TABLE 4. COMPARISON RULES. A KNOWN MISMATCH AND A MISSING BINDING HAVE DIFFERENT CONSEQUENCES.

Requirement	Condition and resulting license
Descriptive comparison	Align target, receiver permission, metric meaning and cost object. Different systems may be described together; token payload and wire bytes cannot be ranked as the same cost.
Paired effect	Compare the same defended test units and outcome. A known test-unit mismatch makes that subtraction incompatible; an unverified fit binding leaves the effect unresolved.
Deployment transfer	Establish a bridge for the changed receiver, intervention, workload, topology or hardware. A demonstrated violation blocks the proposed transfer; an untested bridge leaves it unresolved. Permit only the range and endpoints actually covered.

TABLE 5. FILLED RECORD AND ITS LICENSED JUDGMENT. VALUES ARE SOURCE-REPORTED; NO CROSS-MODE EFFECT IS COMPUTED.

Join key / field	Entry
Family / result	Dangol workflow / `DANGOL-livebinding`.
Version / locator	arXiv:2606.07150v3, §9.7, live-protocol paragraph after Table 5 [16].
Target / view	Workflow classification from live Oblivious HTTP (OHTTP)/Agent-to-Agent (A2A) traffic; not GUI-state observation.
Intervention	Live fixed-rate cover with fresh per-session identifiers, distinct from transformed stored captures.
Workload / testbed	Workflow classes with client, relay, gateway and agent co-located on one loopback host.
Outcome / cost	Accuracy 0.30; about 13× bytes and 11× latency in that condition.
Unresolved fields	Objective terminal task utility U; precise paired live-refit comparison ?.
Licensed judgment	These costs describe this live protocol condition. They neither price dummy GUI actions nor prove task preservation. A matched objective endpoint is needed for the latter.

tion family (Table 9).

4. Corpus Map Across 82 Study Families

4.1. Generating layers, receiving projections and threat polarities

Table 6 maps all 58 core families by signal-generating layers, receiving projections and threat polarity. Multi-layer families retain every reported layer in their generating and observing sets.

Cross-layer synthesis. A remote network observer, a same-site script and a privileged diagnostic hook expose different views even when they share an attribution target. WRA's read-only privacy decoys, AHB's stateful evasion and SplitDecoy's exposed return path illustrate why receiving permissions and protected objectives must accompany the generating layer [6], [14], [34].

Layer incidence highlights where the systematization's comparison problem arises. In the 58-family map, signal-generation labels cluster at P2 serving/runtime (32 families) and P5 orchestration (19); receiving views cluster at P4 application/API (18), P1 hardware (16), and P3 transport (14). Generation at P3/P4 is rare (one family each), despite frequent observation at those layers. This source/receiver asymmetry makes the transfer question concrete: protection at a frequent source layer cannot be presumed to cover a different receiving projection. The map records layer sets, so it does not assign a causal edge to each source/receiver pair.

Purpose labels further show why a single "agent side channel" category is insufficient: Type I appears 41 times, Type II 13, Type III 2, and Type IV 5. The 32 map-only families extend coverage beyond mitigation comparisons: 15 list serving origins, 11 hardware and 11 orchestration; 19 concern confidentiality, nine attribution, one evasion and all five authorized-diagnostic families. Mapping these families locates privacy targets, operator capabilities and candidate validation environments even when no scoped mitigation record enters the ledger. Counts are overlapping family-label incidences; the audit then applies its reuse rules to the 39 scoped result records.

4.2. Relationship to prior systematizations

Prior surveys map compound-system threats, TEE boundaries and privacy contracts [66], [67], [68]. WF already supplies feature leakage analysis, cost trade-offs and environment-shift tests [69], [7], [70], [71]. Mathews et al. re-evaluate nine defenses with stronger attacks and an on-network DynaFlow implementation [8]. Adaptive attacks and online-cost evaluation are established foundations.

What this synthesis adds. Our unit is a *scoped result*, so execution mode, protected state and cost remain attached when one paper reports several mechanisms. The 39-record census exposes these within-paper substitutions; the 82-family holdout matrix separates evidence for each transfer axis. The empirical additions are calibration-source and training-sensitivity measurements, the cost/residual-inference boundary of write-local shaping, and fresh planner behavior after executed responses are withheld by deadlines. WF packet-shaping costs alone do not measure these tool-state and replanning outcomes.

5. Comparability Audit Across All 58 Core Families

5.1. Unified 58-core disposition and 39-record ledger

The corpus partitions as $82 = 58$ core $+24$ adjacent families. Within the core, $58 = 20$ mitigation $+6$ contextual $+32$ map-only families. The ledger holds 39 records: 33 from the 20 mitigation families and one from each contextual family—SIGLEAK, AgentTell, FP-Agent, Behind Bars, LLMs Have Rhythm and ML privacy channels [57], [64], [65], [23], [36], [44]—which supply fixed-guardrail, access-control, interference, VPN or filter-origin contrasts rather than an audited mitigation claim. The 32 map-only families comprise 22 without a scoped mitigation evaluation, five diagnostics, four unbound quantitative assertions and SplitDecoy's boundary case. Table 7 groups mechanisms; its eight rows are mechanism groups, not sampled studies.

Whole-ledger synthesis. Each record has one primary measurement mode: 20 executed collections/prototypes, six trace transformations, five simulations/derived estimates, six component/calibration measurements and two unresolved placements (Table 17). Eleven of the 20 mitigation families contribute more than one record, and nine of them span more than one mode; this is why a mechanism name cannot carry utility or cost between records. Executed collection establishes its stated execution scope, not automatically full-workflow utility.

TABLE 6. COMPLETE MAP OF THE 58 CORE STUDY FAMILIES. EACH FAMILY APPEARS ONCE; LAYER AND TYPE LABELS MAY BE MULTIPLE AND DO NOT IMPLY ONE-TO-ONE CAUSAL PATHS. THE BLOCKS GROUP SOURCE LAYERS, NOT EVIDENTIAL QUALITY.

Core family	$G \to O$	Type
hardware generation, P1 (8)		
Edge EM leakage [19]	$1 \to 1$	I
Energon [20]	$1 \to 1$	II
MoEcho [21]	$1 \to 1$	I
I Know What You Said [22]	$1 \to 1$	I
Behind Bars [23]	$1 \to 1$	I
PAGE404 [24]	$1 \to 1$	I
Spill the Beans [25]	$1 \to 1$	I
ArchSnoop [26]	$1 \to 1$	II
hardware and service, P1/P2 (8)		
SparLeak [27]	$1,2 \to 1$	I
Shape and Substance [28]	$1,2 \to 1$	I
TDXRay [29]	$1,2 \to 1$	I
SparSEEty [30]	$1,2 \to 1$	I
OTRO [18]	$1,2 \to 1$	I
Detokenization Leaks [31]	$1,2 \to 1$	I
LLM-WFIN [32]	$1,2 \to 1,3$	I
LLM-serving RCA [33]	$1,2 \to 2$	IV
service generation, P2 (13 of 22)		
SplitDecoy [34]	$2 \to 2$	I
KVGov [35]	$2 \to 2,4$	I
LLMs Have Rhythm [36]	$2 \to 3$	II
Remote timing [37]	$2 \to 3$	I
NetEcho [38]	$2 \to 3$	I
Prompt Echo [39]	$2 \to 3$	I
Speculative decoding [40]	$2 \to 3$	I
Remote keylogging [1]	$2 \to 3$	I
Whisper Leak [41]	$2 \to 3$	I
Cache contention [42]	$2 \to 4$	I
Routing metadata [43]	$2 \to 4$	I
SafeKV [17]	$2 \to 4$	I
ML privacy channels [44]	$2 \to 4$	I

Core family	$G \to O$	Type
service generation, P2 (9)		
Early Bird [3]	$2 \to 4$	I
Prompt-cache audit [45]	$2 \to 4$	I
InputSnatch [46]	$2 \to 4$	I
Leaky Language Models [47]	$2 \to 4$	I
PromptPeek [2]	$2 \to 4$	I
SpliceLeak [48]	$2 \to 4$	I
Time Will Tell [49]	$2 \to 4$	I
CachePrune [50]	$2 \to 4$	I
MoE prompt stealing [51]	$2 \to 4$	I
cross-layer sources (5)		
ClawGuard [15]	$1,5 \to 1$	IV
MIRAGE [52]	$2,5 \to 2$	IV
Cloud traffic benchmark [53]	$2,5,8 \to 3$	I, II
Web-agent unmasking [54]	$3,7 \to 3,7$	II, III
LumiMAS [55]	$4,5 \to 4,6$	IV
orchestration/workflow sources (15)		
AgentPrint [4]	$5 \to 3$	I, II
Cycle detection [56]	$5 \to 4,6$	IV
SIGLEAK [57]	$5,6 \to 4,6$	I
ProcGrep [58]	$5,6 \to 4,6$	II
AgentProv [59]	$5,6 \to 6$	II
Lidar [60]	$5,6 \to 6$	II
Dangol workflow [16]	$5,6,8 \to 3,6,8$	I
Campaign attribution [61]	$5,6,8 \to 6,8$	II
WRA [6]	$5,7 \to 3$	I
PersonaFingerprint [62]	$5,7 \to 3$	I
Whose Agent? [63]	$5,7 \to 7$	II
AgentTell [64]	$5,7 \to 7$	I
UI Traces [5]	$5,7 \to 7$	II
FP-Agent [65]	$5,7 \to 7$	II
AHB [14]	$5,7 \to 7$	III

G denotes signal-generation layers and O the receiving observation layers. Layer indices: 1 hardware; 2 serving/runtime; 3 transport; 4 application/API; 5 orchestration; 6 tools; 7 browser/environment; 8 multi-agent communication. Types: I confidentiality/privacy; II attribution; III human–agent distinction/evasion; IV authorized diagnostics. These are objectives and observer roles, not mutually exclusive security guarantees. Privileged operator receivers and physical receivers are marked by the layer they inspect. Source versions, locators and normalization of prose-only layer descriptions are in the artifact.

5.2. Within-family and within-cohort record comparisons

The completed audit covers all $\binom{39}{2} = 741$ pairs. The 608 pairs outside the declared mechanism-question cohorts are not comparison failures. The 133 related pairs comprise 66 traffic-envelope, 45 cache-service, 15 local-memory, three behavioral-humanization, three generator-fingerprint and one fixed-instruction pair; they include all 15 same-family pairs. Three findings follow:

- **Same paper, different object.** Eight of 15 same-family pairs substitute primitives, targets or cost objects. CachePrune's direct recovery and contextual guessing differ despite sharing a defense configuration; OTRO's measured unpadded latency does not measure its epoch-padded operating point.
- **Aligned outcome, unpaired population.** Of the other seven same-family pairs, four have different collection/execution populations and three lack a verified matched-test binding. Dangol's stored-capture ablation and live loopback remain distinct deployment measurements.
- **Across records.** Of 133 related pairs, 17 (12.8%) support a source-qualified descriptive comparison, 105 (78.9%) have an established object/receiver/target mismatch, and 11 (8.3%) remain unresolved. Descriptive compatibility permits scoped comparison, not pooled effects or a ranking across heterogeneous costs.

The final inventory retains all 528 earlier pair judgments and adds 31 related judgments for the six expansion records, with source locators and reasons. Examples illustrate this census rather than define its sample.

Why the record changes the reuse decision. The census distinguishes three outcomes for a deployment query. A known object mismatch requires a new measurement at

TABLE 7. EIGHT INTERVENTION FAMILIES ORGANIZED BY THREAT POLARITY (TYPE I: CONFIDENTIALITY PROTECTION; TYPE II: PROVENANCE/IDENTITY ATTRIBUTION; TYPE III: BOT-DETECTION EVASION), RECEIVING VIEW AND SUPPORTED DEPLOYMENT JUDGMENT. BECAUSE OBJECTIVES AND RISK MAGNITUDES DIFFER, THESE ROWS ARE NOT A COMMON EFFICACY RANKING.

Intervention & polarity	Evaluated view, target and source	Established result and applicable boundary
Padding / batching (Type I: confidentiality)	Network packet lengths and groups; topic inference. NetEcho and Whisper Leak [38], [41].	Collected padding traces retain inference signal; packet-group batching is evaluated by replay. A protected upstream length view does not cover downstream re-emission or residual grouping. Live incremental cost remains U.
Constant-rate / transport cover (Type I: confidentiality)	Network workflow fingerprint; Dangol's OHTTP testbed [16].	The co-located loopback setting yields 0.30 recovery at about 13× bytes and 11× latency. Transformed-capture results are separate; neither result supplies objective agent-task utility.
Concurrent query decoys (Type I: confidentiality)	Destination/domain observer; private prompt and trait inference. WRA [6].	Real/decoy research prompts execute in parallel as read-only queries whose decoy answers are discarded locally; interleaved-domain evaluation reduces one-decoy E_{func} from 0.770 to 0.610. Terminal decoy utility/cost are U.
Action humanization (Type III: bot-evasion)	Mobile GUI touch traces; evading human–agent bot detection. AHB [14].	Unlike WRA's read-only privacy decoys, AHB's `Fake` mode dispatches live mobile GUI touches to evade platform bot detectors, dropping Trip Planning success from 0.75 to 0.15. Non-Fake `History+Long` (0.7143 success) avoids GUI mutation.
Cache isolation / keying (Type I: confidentiality)	Co-tenant prefix timing; SafeKV, KVGov and CachePrune [17], [35], [50].	SafeKV reports post-trigger AUC near 0.5 and separate TTFT/throughput comparisons. KVGov's HMAC median 1.3 μs prices key computation, excluding lost reuse. CachePrune jointly binds marked-token protection, QA/summarization utility and serving performance.
Cache timing equalization (Type I: confidentiality)	Co-tenant cache-fusion timing; SpliceDefense [48].	Constant-Time Boundary Fusion (CTBF) suppresses the tested signal over 15 tasks/75 steps. Quantized Chunk Padding (QCP) adds mean 7.5 tokens and $<$ 5ms/$<$ 1% prefill time; this does not price combined deployment.
Timestamp perturbation (Type II: attribution)	Recorded website action metadata; model/agent attribution. UI Traces [5].	Fitting on delayed traces (D_2) recovers substantial attribution performance under offline timestamp edits that retain action types/order. Live execution delay and task outcomes are not measured by the trace transformation.
Oblivious / unshared memory (Type I: confidentiality)	Host or local resource observations; TDXRay, Gao et al. and OTRO [29], [22], [18].	TDXRay measures oblivious-map lookup cost; Gao measures loading/memory cost after disabling zero-copy. OTRO reduces conditional access-count information (3.94 to 0.04 bits) with separate unpadded TTFT (4.59%) and epoch-padding (2.8ms) bounds.

Authorized operator diagnostics (Type IV: ClawGuard, cycle detection, LumiMAS, LLM-serving RCA and MIRAGE) are baseline telemetry rather than privacy attacks or evasion defenses, and are cataloged separately.

the intended operating point; an unresolved binding requires source reconciliation; a descriptive match permits a scoped comparison while retaining each cost's denominator. These outcomes prescribe different work. For example, aligning two packet-size accuracies cannot supply the missing online execution cost, and measuring a component's throughput cannot establish a planner's success under response deadlines. The 17 descriptive matches are useful precisely because their permissions remain explicit.

5.3. Nine-item reuse audit across 20 mitigation families

R1–R9 are author-defined reuse conditions, not a quality score. Table 8 links requested conclusions to necessary evidence. Missing evidence can block an analyst's deployment extension while leaving the source's own claim intact. Component utility suffices for component claims; R5 applies to adaptation comparisons. U and ? retain distinct source states: the 11 unverified cells are five R4 and six R5 bindings, which are distinguished from an attack failure.

Join evidence at the record, not the family. A family-level B means that at least one record supplies an item. It does not imply that its other B cells concern the same intervention. We therefore code all 351 record–item cells and join only B bindings within the same record and configuration (Table 10). Denominators exclude a record if any required item is inapplicable; U, ?, and P remain eligible but unbound. The six contextual controls are separate. Only CachePrune supports the 2/20 join: two outcomes share one quality/cost package, not two independent confirmations. The denominator identifies eligible records for our query, not papers obliged to answer it. SparLeak's matched ASR/ROUGE pairs qualify as component utility under R8, correcting its former partial code.

Which evidence packages are available. Record-local joins resolve progressively stronger deployment queries: inference with utility is bound in 4 of 22 eligible records (three families/configurations), inference with cost in 5 of 22 (three families, four configurations), and matched adaptation with inference in 2 of 17. Their intersection cannot be reconstructed by adding these counts: only CachePrune's two outcomes share one inference–utility–cost package. A deployment analyst can therefore reuse the bound component outcome, price a missing cost, or measure workflow utility according to the actual join. This is more informative than treating every family-level B as a complete deployment result.

SoK Takeaway 2 (Join the operating point). CachePrune's two outcomes identify one reusable inference–utility–cost package. Request evidence for the intended deployment query rather than combine a family's best individual cells. These counts inventory evidence; they do not rank paper quality.

TABLE 8. CLAIM-CONDITIONED REUSE RULES. EACH ROW NAMES THE REQUESTED INFERENCE AND THE EVIDENCE NEEDED TO SUPPORT IT.

Item	Reuse query → required binding (**B**)
R1 Target	Same protected interest → explicit hidden property, inference goal and authorization relation.
R2 Receiver	Same observer protection → permissions, released view and auxiliary inputs.
R3 Intervention	Same executed mechanism → placement/mode: stored data, wire emission or actions.
R4 Protected fit	Resistance to protected-data fitting → identified protected training/calibration and testing.
R5 Matched adaptation	Effect of adaptation → clean/protected fits tested on the same defended population.
R6 Workload	Changed workload/system → tasks, configuration and the relevant holdout axis.
R7 Inference	Quantified protection → metric, denominator and scoped defended evaluation condition.
R8 Utility	Preserved task/component function → performance at the same protected state/operating point.
R9 Cost	Resource-budget feasibility → units, baseline and cost object at that operating point.

B: bound; **P**: partial binding; U: unreported in inspected scope; A: source unavailable; ?: binding unverified; –: inapplicable. First determine applicability, then assign evidence status. Family **B** means at least one scoped record satisfies the item, not that every result does.

TABLE 9. PER-FAMILY EVIDENCE BINDINGS FOR ALL 20 EVALUATED-MITIGATION FAMILIES. R1–R9 ARE SCOPED REUSE ITEMS, NOT OVERALL PAPER SCORES.

Family	R1	R2	R3	R4	R5	R6	R7	R8	R9
UI Traces [5]	**B**	**P**	**B**	**B**	**B**	**B**	**B**	U	–
WRA [6]	**B**	**P**	**B**	–	–	**P**	**B**	**P**	U
NetEcho [38]	**B**	**B**	**B**	**B**	**P**	**B**	**B**	U	U
Early Bird [3]	**B**	**B**	**B**	**B**	**P**	**B**	**B**	U	**B**
Whisper Leak [41]	**B**	**B**	**P**	?	?	**B**	**B**	U	**B**
SpliceLeak [48]	**B**	**B**	**B**	–	–	**B**	**B**	U	**B**
Time Will Tell [49]	**B**	**B**	**B**	?	?	**B**	**B**	U	**B**
I Know What You Said [22]	**B**	**B**	**B**	–	–	**B**	U	–	**B**
Speculative decoding [40]	**B**	**B**	**P**	?	?	**B**	**B**	–	**B**
Dangol workflow [16]	**B**	**B**	**B**	**B**	**B**	**B**	**B**	U	**B**
Cloud traffic benchmark [53]	**B**	**P**	**B**	?	?	**P**	**B**	**P**	U
Remote timing [37]	**B**	**B**	**B**	–	–	**B**	**B**	U	**P**
TDXRay [29]	**B**	**B**	**B**	–	–	**B**	U	**B**	**B**
SafeKV [17]	**B**	**B**	**B**	–	–	**B**	**B**	U	**B**
AHB [14]	**B**	**B**	**B**	**P**	?	**B**	**B**	**B**	U
KVGov [35]	**B**	**B**	**B**	–	–	**B**	**B**	U	**B**
CachePrune [50]	**B**	**B**	**B**	–	–	**B**	**B**	**B**	**B**
OTRO [18]	**B**	**B**	**B**	–	–	**B**	**B**	–	**P**
SparLeak [27]	**B**	**B**	**B**	?	?	**B**	**B**	**B**	U
Shape and Substance [28]	**B**	**B**	**P**	U	U	**B**	**P**	U	**P**
Bound	20	17	17	4	2	18	17	4	11
Applicable	20	20	20	11	11	20	20	17	19

B: bound; **P**: partial; U: unreported in inspected scope; ?: unverified binding (11 cells); –: inapplicable. Table 8 defines R1–R9 and the shared status alphabet. A bound cell is supported by at least one scoped record. Dangol R5 concerns stored capture; its live-refit binding remains unverified. CachePrune jointly binds R7–R9 at one operating point.

TABLE 10. CONFIGURATION-LINKED EVIDENCE FOR REUSE QUERIES; CONTEXTUAL CONTROLS EXCLUDED. COUNTS DESCRIBE THIS COLLECTION, NOT PAPER QUALITY OR FIELD PREVALENCE.

Binding	Records	Families	Configs
Inference + utility	4/22	3	3
Inference + cost	5/22	3	4
Matched adaptation + inference	2/17	2	2
Inference + utility + cost	2/20	1	1
All five bindings	0/14	0	0

Joins link configurations, not necessarily identical requests or paired effects. The 2/20 row is CachePrune alone: two outcomes, one family/configuration. Eligible sources need not claim the combined result. Cost retains its payload/capture/wire scope. The supplement lists all 351 cells and positive identities.

6. Systematic Holdout Evidence for Transfer

6.1. Six axes across all 82 families

Appendix D gives the complete family-by-axis matrix, including the 24 adjacent families. It codes task/input/user, model, harness/topology, hardware, network/load and time separately from the archived source-bound extraction. H denotes an explicitly reported held-out unit/domain; V a condition comparison or recalibration; X a scoped restriction; ? unresolved extraction. H does not certify every grouping detail or successful transfer, and ? is never counted as absence. Conceptual families receive an inapplicable dash.

The matrix contains H entries for 13 families on task/input/user, three on model, two on harness/topology and one on network; none is resolved as H for hardware or time. The latter axes contain seven V entries each: calibration, condition sweeps and temporal observations already exist in the literature. With 73 hardware and 71 temporal entries unresolved, this is a map of *documented evidence*, not a claim that those papers omitted such tests. Multiple H entries are counted once per family/axis, not pooled as independent experiments.

6.2. What each holdout licenses

Whisper Leak holds out target prompts; FP-Agent has a separate task holdout but unresolved visitor/host separation. Dangol's leave-composition-out and the traffic benchmark's topology shift change a different generator [41], [65], [16], [53]. LLM-Fin and ArchSnoop test unseen variants/configurations within known model families; LeakyLMs conditions architecture inference on known GPU type and calibration [72], [26], [47]. Thus a positive model holdout does not establish unknown-hardware transfer. UI Traces' recorded question overlap further shows why task labels and actual held-out units must remain distinct [5].

6.3. Operational consequence

Select the axis changed by deployment, require a source-bound separation on that axis, then check whether fitting or calibration crossed the split. Otherwise scope the result to its observed condition. Feature ablations and additional models alone supply V evidence, not a bridge to an unseen generator. This rule also governs our endpoint exper-

iments: within-model refits, source-trained attacker transfer and source-calibration transfer occupy separate matrix cells. The systematic result is the mismatch between the requested transfer axis and the available evidence, not a ranking by the number of holdouts.

From holdout counts to a transfer obligation. Each axis names a distinct source of variation. A task holdout tests new inputs under its stated generator; a model holdout changes the generator; topology and network holdouts change the receiving path; hardware and time can change the implementation or scheduling process. The 13 task/input H entries and three model H entries therefore answer different questions, rather than form a single robustness score. Where extraction remains unresolved, the next action is to establish the source's scope before demanding a new experiment. Where a deployment changes several axes, evidence must cover their intended combination: separate model and network tests do not automatically establish their joint operating point.

7. Synthesis: From Evidence to Reuse Decisions

The census (§5) covers all 39 records and the holdout matrix (§6) all 82 families. We now select three examples by the *comparison decision* they expose, not by effect size: a changed receiving path, cover with different state effects, and an adaptation/cost mismatch within a named mechanism. They explain mechanisms within the complete census and motivate the selected controlled measurements in §8.

7.1. Case 1: bind the final target-bearing emitter

NetEcho's upstream packet-size shaping operates on a different projection from the tool-call sequence. Its plaintext-aware frontend then emits a new size trace (grouped token counts or character lengths), reopening a size channel after upstream protection [38]. SplitDecoy is a separate return-path record: the cloud sees nonzero gradient row support on every tested frame across nine seeds [34, Sections 5.1–5.3]. Both change the receiver boundary and therefore require a separately bound inference result.

The data processing inequality requires $S \to Z \to Y$: target-oblivious postprocessing cannot increase $I(S; Z)$ [73, Theorem 2.8.1]. Plaintext-aware re-emission or backward computation breaks that premise. Exp 1 measures what restoring it costs under buffered and streaming emission.

7.2. Case 2: resolve the WRA versus AHB cover asymmetry

WRA's one-decoy E_{func} drops from 0.770 to 0.610, whereas AHB's Trip Planning success falls from 0.75 to 0.15 under `Online+Fake` humanization [6], [14]. Grouping both as "cover traffic" conceals different objectives and executions.

WRA discards decoy reads for Type I confidentiality; AHB executes GUI events for Type III evasion, potentially navigating or submitting forms. Their utility outcomes are not contradictory, and AHB's missing condition-specific denominators preclude a matched causal effect. A shared workflow setting does not establish shared state effects. Terminal-state reward and dispatch-time authorization also answer different questions: an unconfirmed write can still reach the intended final state. Exp 2 checks both.

7.3. Case 3: bind adaptive retraining and cost objects

A matched adaptation contrast compares clean-trained D_0 and defense-trained D_2 on identical defended tests. UI Traces and Dangol raise the record-binding question this leaves open: whether a refit, a measured delay and a utility outcome refer to the same executed condition [5], [16]. Cost objects show the same pattern. Early Bird's 6ms/4% GPTCache cost does not price K-prefix isolation; KVGov's $1.3\,\mu$s HMAC microcost omits lost reuse; SpliceDefense's QCP cost does not price CTBF+QCP [3], [35], [48]. CachePrune binds protection, QA/summarization utility and throughput, while OTRO separates conditional leakage from TTFT/padding costs [50], [18]. Exps 3–4 test calibration, refitting and timing against one cohort and matched observation sets.

7.4. A deployment query determines the evidence to retain

Start with the intended receiver, protected outcome and intervention. Use the pair audit to check that the cited result measures that object, the record join to identify which inference, utility and cost outcomes share its configuration, and the holdout matrix to locate the deployment change. This sequence preserves usable evidence while identifying the specific missing measurement. Table 11 makes the result operational: a new emitter needs a receiver-specific capture, writes need state/authorization checks, a new calibration source needs a local recalibration baseline, and an online scheduler needs its own planner utility and latency measurement.

The following studies instantiate these decisions at selected operating points. Their role is to measure the consequences and trade-offs of a changed boundary. The systematized conclusion rests on the complete literature audit; the experimental numbers retain their named workload, receiver and fitted attack suite.

8. Controlled Studies Supporting the Synthesis

Four studies attach measured consequences to the evidence-reuse decisions of §7: emitter, executed state, calibration and attacker observation. Exps 1–2 reproduce mechanisms and measure their costs; Exps 3–4 examine agent operation signatures, training variability and execution

TABLE 11. DEPLOYMENT CHANGE → MISSING BINDING → MINIMUM DECISION EVIDENCE. EACH ROW NAMES A MEASURABLE GATE FOR CARRYING A RESULT FORWARD.

Changed boundary	What the old result leaves open	Minimum comparison before reuse
Emitter / representation	Buffered or upstream cost and inference do not price a re-emitting/streaming endpoint.	Capture the final receiver; fit its attacker; report overflow, bytes and first-output/completion delay (Exp 1).
Read cover → writes	Discarding a reply does not undo commits, refunds or authorization effects.	Compare live database/authorization outcomes; execute cover on the chosen isolation path; include its calls and quotas (Exp 2).
Customer / model calibration	A template's target-wise inference and overhead can change with the receiving trajectory distribution.	Hold out customers; compare imported calibration with resampled local calibration; refit the full attack suite and correct target-wise comparisons; report byte/call ratios (Table 15).
Attacker observation	Chance-level length inference does not bind timing or flush identification.	Re-evaluate the same captures under a common expanded suite; pair the suite-max change (Exp 4: cancellation $+11.4$ points, interval $[3.7, 19.6]$).
Loopback → WAN	A fixed tunnel horizon does not guarantee completion under network delay.	Measure completion/overflow under the target path; retune the horizon on training conditions and reprice protection (Exp 4).
Replay → live planner	Preserved logical responses do not measure timeout/replanning utility.	Run fresh paired planner episodes with the actual scheduler, delays and deadlines; measure success and latency. Prefix-cover utility is measured here; full framing/flush needs its own run.

TABLE 12. COHORT, OBSERVATION AND COST DENOMINATORS FOR EXPS 3–4. STANDARD-PROMPT PAIRS: CANCEL/RETURN BA (%). BYTE RATIOS PRICE JOINT SHAPING AGAINST EACH STUDY'S OWN UNDEFENDED SERIALIZATION.

Study	Observation / attack suite	Undef. BA	Joint B×
Exp 3 frozen	Replay lengths/order; frozen suite	97.2/94.0	15.39
Exp 3 cross-fit	Analytic lengths/order; 11-attack union	91.7/85.5	19.76
Exp 4	New RPC envelope; lengths/order/time; full union	91.7/86.4	15.11

Frozen-split and Exp 4 rows: 276 test episodes, 24 customers, 5/7 positive customers. Exp 3 cross-fit: 690 episodes, 53 customers, 14/18 positive customers. Each prompt has half the episodes. Cross-fit fold calibration yields larger templates ($5.04\times$ calls); frozen-split and Exp 4 rows use $3.83\times$. Repeats measure execution variation within these customer groups.

conditions. Populations remain distinct (Table 12). Supplement S31 indexes the detailed operating grids; Appendix C indexes the studies.

Measurement contract. A transparent relay captures post-handshake TLS application records on loopback. BA averages class recalls; reported BA is the maximum over the named attack suite. Intervals reselect that maximum over 2,000 task- or customer-cluster bootstrap draws. The frozen size/order suite uses clean-trained and defended-refit learners plus decoder/count rules; the endpoint suite adds total-exchange count. Cross-fit/localization use an 11-attack union; timing uses a 17–22-attack union, with seven shared learners for offline controls. R21 additionally resamples calibration and attacker training. The protocol's absolute screen (upper 95% BA below 0.60 and AUROC below 0.65 for both targets) remains unmet in Exps 3–4; the comparisons below measure scoped inference and cost.

8.1. Exp 1: emitter changes require new inference and cost evidence

For 48 saved brief/expanded responses on 24 topics (12 training, 12 test), padding only the upstream link leaves verbosity 100% separable after plaintext-aware re-emission, reproducing NetEcho's mechanism [38]. A buffered training cap reaches 50% BA at $2.18/2.19\times$ bytes under TLS 1.2/1.3. Streaming the same texts per token changes both the observable and cost: eight-token batching gives 50% clean-trained BA but 100% after defended refitting; constant event size and count reach 50% at $2.42\times$ bytes and 208 versus 105 token intervals, with two test overflows [1], [41]. **Reuse decision:** a buffered cap cannot price streaming protection; capture and fit at the final emitter, including overflow and delivery delay.

8.2. Exp 2: read cover does not bind write-state preservation

Across 97 initially eligible retail tasks and three seeds, same-item modification changes status and appends a refund in 173 of 291 placements; 118 fail availability checks. An idempotent operation can still change the original state. Shadow execution isolates cover effects and suppresses responses; the live-cancellation control changes the database in every case. These conformance checks establish the chosen state boundary.

On all 115 reference programs, task-folded count/order/length attacks recover 100% BA after refitting to a single cancellation decoy. Equalizing the target count leaves 76.9%; equalizing the full tool shape reaches 50.0/54.5% cancel/return BA at 6.71 extra calls per real call. The 4,140-session TLS composition test then shows why dispatch and transport are separate: per-call buckets leave return BA at 73.9%, whereas joint cover/framing reaches 54.4/50.0% at $26.3\times$ bytes and $7.7\times$ calls. **Reuse decision:** bind cover to its executed state and observed link; a passive-link tunnel can replace tool-executing cover

TABLE 13. FROZEN EXP 3 COMPARISON, STANDARD PROMPT: 138 TEST EPISODES, 46 TASKS, 24 HELD-OUT CUSTOMERS. BA (%) USES THE FROZEN SUITE; BRACKETS GIVE CUSTOMER-CLUSTER 95% INTERVALS.

Condition	Cancel BA	Return BA	B×/C×
Undefended	97.2 [90.0,100.0]	94.0 [85.8,100.0]	1.00/1.00
Padding only	66.1 [50.0,83.9]	55.5 [50.0,73.0]	4.02/1.00
Shape only	60.8 [55.4,71.3]	51.9 [50.5,57.8]	4.16/3.83
Joint	50.0 [50.0,60.6]	50.0 [50.0,65.5]	15.39/3.83

B/C: byte/call ratios to raw transport. Joint replays the same real-agent decisions with isolated prefix cover and constant framing; planner utility is measured separately. Supplement S31 retains both prompts, original rewards and paired contrasts.

at that link, but dispatch protection after decapsulation requires isolation (§9.1).

8.3. Exp 3: separate operation inference, calibration and utility

Frozen operating point. GPT-6.1 Sol/low executes 115 retail tasks with two prompts and three repeats: 690 runs from 53 customers. Targets are the requested cancellation/return operations in the reference program. The rules prompt adds reads before order access and writes. All 414 training runs precede freezing cover donors, templates and framing caps; 276 test runs cover 46 tasks and 24 disjoint customers. Prefix cover flushes unmet counts at episode end. Replaying each trajectory under four program conditions per prompt yields 2,760 local TLS captures; real responses and live state are preserved by construction (Appendix A). Rules improve read-order adherence from 86.2% to 97.1% while undefended inference remains above 93%: policy compliance and operation privacy separate. Joint shaping reduces standard-prompt BA from 97.2/94.0% to 50.0/50.0%; upper bounds are 60.6/65.5%. Exploratory customer-grouped cross-fitting over the full known cohort, with all calibration and 11 attacks refit per fold, yields 60.5/55.5% at 19.8× bytes and 5.0× calls. The scoped cost and fitting regime accompany each result.

Localization and model-specific evaluation. Writes are 19.6% of real calls yet retain oracle-selected 91.7/84.9% BA; removing them leaves 62.8/59.9%. This locates the metadata signature of an operation-defined target and motivates testing write-local intervention. On MiniMax/GLM, write-only padding costs 1.45/1.51× bytes but leaves return BA at 61.3/61.6%. A learned write detector has precision 34.9/98.3% and recall 96.3/89.4%, so oracle localization and implementable inference have different scopes. Luna contributes a 150-run same-provider extension; MiniMax and GLM each contribute 115 runs, a 69-task training calibration frozen before 46 test tasks, and the complete endpoint suite (Table 14). These are within-model comparisons on the known retail cohort. MiniMax is a served alias with an operator-reported DeepSeek backend; GLM's token budget is 16,384 versus MiniMax's 4,096. Neither exposes an immutable model revision.

TABLE 14. EXP 3 MODEL-SPECIFIC REPEATED EVALUATIONS: NETWORK SUITE-MAX BA (CANCEL/RETURN, %). EACH MODEL HAS 46 TEST TASKS AND 24 CUSTOMERS. ENDPOINT ROWS USE THE ENDPOINT SUITE; LUNA RETAINS THE FROZEN SUITE. LUNA USES 29 TRAINING TASKS; THE TWO ENDPOINT REPLICATIONS USE 69. CONTRASTS ARE WITHIN MODEL; ENDPOINT ALIASES AND INFERENCE BUDGETS DIFFER.

Model	Undefended	Joint
Luna	64.2 / 84.9	60.4 / 57.0
MiniMax-M3	58.3 / 83.5	60.4 / 67.5
GLM-5.3-flash	100.0 / 86.6	50.0 / 67.3

Calibration source versus local training variability. Table 15 swaps calibration source and refits the endpoint suite on target training customers. Importing MiniMax calibration changes GLM cancellation/return BA by $+20.8/-17.3$ points and increases bytes. Customer-paired permutation tests do not resolve either direction (raw $p = 0.134/0.174$; Holm-adjusted $p = 1.000$ across eight contrasts). In 100 paired training-customer resamples, local GLM return BA spans 50.0–67.9% (95% sensitivity range); imported-minus-local return has median 0.0 and spread $[-16.9, 18.4]$ points, or $+0.3$ $[-11.2, 18.9]$ with attacker resampling. **Reuse decision:** measure local recalibration and attacker-fit variability before attributing a contrast to model source.

Planner utility at the executed scheduler. A paired 50-task GLM study runs 100 fresh episodes with isolated prefix-cover execution and emulated response deadlines. Baseline/prefix-shadow succeed on 37/50 and 33/50 tasks (difference -8.0 points, customer interval $[-23.9, 6.5]$). Withheld executed-tool responses trigger later planner actions. This measures the prefix scheduler's utility; the replayed joint-framing/terminal-flush mechanism needs its own live utility measurement. **Reuse decision:** retain logical response/state conformance and measure deadline/replanning utility at the actual scheduler.

8.4. Exp 4: observation and path changes alter the trade-off

All 690 saved action trajectories execute through a local tool server under six conditions (4,140 TLS sessions); each real response gates the next action. The original 414/276 split is retained and attackers are frozen before test capture. Serial/two-in-flight shadow schedules use constant per-call framing; full tunnels send 4,096-byte cells in both directions at 2/5 ms intervals over a training-sized 133-slot horizon. Dummy cells execute no tools.

Expand the observation on matched captures. Adding timing/flush attacks to size/order attacks increases serial-shadow cancellation BA from 50.0% to 61.4% (paired $+11.4$ points, interval $[3.7, 19.6]$); return changes from 50.7% to 53.4%. Flush detection reaches 90.4/92.3% for serial/parallel schedules; composing a learned suffix detector with operation inference yields a strongest return point of 60.7%. The full union retains stripped and unstripped traces.

TABLE 15. CALIBRATION SOURCE, LOCALIZED PADDING AND TRAINING SENSITIVITY. TOP: ORIGINAL 46 TEST TASKS / 24 CUSTOMERS PER ENDPOINT; Δ IS THE SEVEN-ATTACK MAX-BA CONTRAST AGAINST TARGET-LOCAL JOINT SHAPING (POINTS). BOTTOM: 100 PAIRED TRAINING-CUSTOMER RESAMPLES, EVALUATED ON THE SAME FIXED TESTS.

Target	Calibration / intervention	Cancel BA	Return BA	Δ cancel	Δ return	Bytes×	Calls×
MiniMax	MiniMax joint	60.4	67.5	–	–	20.06	3.36
MiniMax	GLM joint	50.8	63.0	−9.6	−4.5	17.48	3.11
MiniMax	Write padding	50.0	61.3	−10.4	−6.2	1.45	1.00
GLM	GLM joint	50.0	67.3	–	–	15.95	4.24
GLM	MiniMax joint	70.8	50.0	+20.8	−17.3	18.65	4.66
GLM	Write padding	60.0	61.6	+10.0	−5.7	1.51	1.00

Target	Label	Local recalibration BA: 95% spread	Imported − local Δ: median [95% spread]	Same Δ with attacker resampling: median [95% spread]
MiniMax	cancel	[50.0, 64.8]	+0.0 [-13.0, 11.1]	+0.0 [-14.6, 16.5]
MiniMax	return	[55.2, 73.4]	-4.5 [-20.2, 6.1]	-1.2 [-11.7, 10.8]
GLM	cancel	[50.0, 69.2]	+2.5 [-13.8, 17.1]	+2.9 [-14.4, 17.5]
GLM	return	[50.0, 67.9]	+0.0 [-16.9, 18.4]	+0.3 [-11.2, 18.9]

Original eight customer-paired contrasts: 99,999 whole-suite swaps each, Holm-adjusted $p = 1.000$ throughout. Historical nominal intervals remain in the supplement. Resampling spreads measure training sensitivity, not confidence in a population effect. All seven attacks (including decoding and call counts) are refitted per condition. Costs are summed local TLS test bytes/calls relative to raw target totals; they exclude live planner utility and WAN delay.

TABLE 16. EXP 4 OPERATING DECISIONS: STANDARD PROMPT, 138 TEST EPISODES, 24 CUSTOMERS (5/7 CANCEL/RETURN-POSITIVE). BA PAIRS ARE CANCEL/RETURN (%). M: MEASURED TOOL RPC ON SAVED ACTIONS; S: OFFLINE METADATA TRANSFORMATION.

View	Condition	Common-seven BA	Full-union BA	Bytes×	Calls×	RPC p95 (ms)
M	Undefended	84.3/76.9	91.7/86.4	1.00	1.00	15.4
M	Per-call constant	53.2/59.1	68.6/62.7	3.92	1.00	16.6
M	Serial shadow	50.0/52.9	61.4/53.4	15.11	3.83	26.2
M	Two-in-flight shadow	50.0/57.3	51.7/57.7	15.11	3.83	20.8
M	Full tunnel, 2 ms	59.9/55.2	62.5/55.2	125.71	1.00	56.4
M	Full tunnel, 5 ms	54.6/53.7	54.6/57.3	125.71	1.00	112.3
S	WT random-donor	50.0/56.7	50.0/56.7	2.04	1.00	–

M unions contain 17–22 attacks; S uses the common seven. RPC latency ends at the last real response and excludes LLM generation; S has no measured RPC latency. Supplement S31 retains all 13 conditions and 95% upper bounds, each above 60%. The S row supplies a descriptive shared-suite candidate for online evaluation.

Thus recognizable cover phases and downstream operation recovery are measured separately.

Retain the cost object when comparing mechanisms. Measured RPC conditions show a cost–inference trade-off: per-call constant framing uses 3.92× bytes and 16.6 ms p95 at 68.6/62.7% full-union BA, versus serial shadow's 15.11×, 26.2 ms and 61.4/53.4%. Offline random-donor WT reaches 50.0/56.7% at 2.04× under the common seven, identifying a lower-byte candidate for online evaluation. Table 16 keeps its transformed metadata and simulated span separate from measured RPC latency and the extended timing suite.

Size the transport horizon to the receiving path. Re-running all 276 test episodes under 20/5 and 50/10 ms one-way delay/jitter retains 89.4% flush-detector BA versus 90.4% locally, while serial-shadow cancellation inference falls to 50.0/54.0%. The 2 ms tunnel's 266 ms horizon completes only 85/276 and 5/276 sessions; shadow/per-call framing complete all, with serial-shadow p95 of 855.4/1,999.5 ms. **Reuse decision:** expand the attacker's observation on matched captures, measure online latency for the chosen mechanism, and reprice a horizon calibrated for the target path. Together these studies supply operating examples for the SoK's decision table.

9. Implications, Ethics and Limits

9.1. Deployment paths and reporting obligations

Table 11 specifies the missing measurement for each changed boundary. A passive observer may be an access-network operator, VPC monitor or inter-organization relay. Remote paths expose destinations even with Encrypted ClientHello [74]; our measurements use one ingress endpoint. Host instrumentation and downstream commit logs require different protection.

Shadow state can use authorized reads/verified no-ops, provider isolation (Kubernetes dry-run, Stripe sandboxes, transactional rollback), or a trusted gateway routing marked writes to an isolated executor [75], [76], [77]. Idempotence alone is insufficient, and sequence changes can survive rollback [78]. We measure the gateway path; provider integrations remain designs. The gateway can identify cover and is trusted. Ingress shaping competes with transport controls; restricted dispatch protection additionally requires hidden markers/routing. Quotas, backpressure and separate credentials must prevent overload and live side effects. The supplement details these deployment contracts.

9.2. Ethics and artifact scope

Executions use cloned public benchmark databases and loopback sockets; model requests contain public tasks and simulated interactions. No production account is modified or third-party service probed for a vulnerability. Stakeholders include paying tenants, shared-capacity providers and people affected by state changes. Other-account cancellation is a simulator control. Released inference code can support surveillance; examples use public/synthetic identities and controlled endpoints. Bot-detection evasion has a different authorization boundary [14], [6].

AI tools assisted candidate-field extraction, implementation, source reconciliation and prose revision; the authors verified the original source-linked corpus and retain responsibility for the revised synthesis. The artifact archives source locators, coding decisions, observations, calibration and executable checks.

9.3. Threats to validity and scope of evidence

Statistical scope. The BA/AUROC screen is an engineering threshold; no defended configuration meets it for both targets. Original nominal intervals condition on fitted suite predictions. Calibration contrasts add Holm-corrected permutations and 100 training-customer resamples (Table 15), reusing observed customers. The frozen/full cohorts have 5/7 and 14/18 positive customers; repeats add execution coverage. Real-agent calibration precedes test collection but developers knew the tasks. Reference calibration uses the known population; cross-fit, timing and transfer extensions are exploratory.

Targets and deployment utility. Operation targets are coupled to intended actions; localization identifies their metadata signature. The same-operation attribute diagnostic is inconclusive (Appendix C). Original agent reward, defended logical conformance and live prefix-shadow utility are distinct outcomes. Full framing/flush utility remains unmeasured. Loopback RPC timing excludes LLM generation, and WAN effects are emulated in the relay rather than measured on wide-area paths. WF adapters support matched metadata comparisons; original-protocol equivalence and equal-utility deployment comparisons remain open.

External validity. The four mutable model aliases operate in one retail environment with separate user-simulator contexts. Luna has 29 training tasks; MiniMax and GLM each have 69. Transfer matrices and live planning reuse known customers. Fresh customers, fixed weights, verified cross-provider identities, new environments and provider-backend measurements define the next validation axes.

10. Conclusion

This SoK organizes defense evidence around the deployment change for which it will be reused. The complete record audit exposes within-family object substitutions; configuration-linked joins and the six-axis holdout map identify the evidence that remains applicable. Controlled studies attach inference, state, utility and cost consequences to selected changes, including local calibration variability and timing-aware evaluation. The resulting decision table specifies which measurement is needed before carrying a defense claim to a new operating point.

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Appendix A.
Audit Trail and Logical Equivalence

Evidence organization. The package separates the 82-family source map, 39-result ledger, 180-cell audit, transport tests, dispatch tests and real-agent studies. Each source

record has a version and section/table/figure/paragraph locator. The supplement supplies detailed algorithms, full result grids and archived coding decisions; the main paper contains the comparison rules, complete family matrix and record index needed to follow the synthesis.

Logical equivalence. For deterministic tools, a real action gives $(s_{t+1}, o_t) = T(s_t, a_t)$. The defended dispatcher forwards that action unchanged, runs cover on a disjoint state s_t^{sh} and suppresses cover responses. With equal initial states and realized real actions, induction gives

$$s_t^D = s_t, \qquad \pi_{\mathrm{plan}}(D(H_t)) = H_t.$$

Here H_t is the logical message/action/response history, excluding wall-clock time. Payload-preserving framing and terminal shadow flush maintain the invariant, provided timeouts, shared-state side effects and timing-dependent tool behavior are absent. Checks compare every real-call projection and scoring field across all 690 trajectories and four program conditions. These establish implementation conformance to isolation. The utility boundary in §9.3 explains the further live-planner measurements.

The reproducibility index resolves each archived experiment in Appendix C to protocol, inputs, predictions and checksums. Saved-output verification reproduces the reported metrics without new model calls, while measured captures and analytic reconstructions retain their separate provenance.

Appendix B. Result-record index

Table 17 indexes all 39 records with their primary measurement mode. Separate records identify changes in observation, intervention, target, execution population or cost object within a family. The six contextual records (‡) and the six records added by the final search (†) are marked; the remaining 27 are the mitigation records of the initial ledger, expanded to all 39 records in the final pairwise audit (§5). Shortened source locators support direct lookup; the machine-readable ledger retains exact versions, hashes and full locators.

Appendix C. Experiment artifact index

Table 18 maps the four main experiments to stable artifact keys and detailed supplementary methods. Reanalysis, prompts and repetitions retain their original customer/task membership.

The reproducibility index resolves these keys to files, protocols, saved predictions and executable verification commands. The source versions and calibration chronology remain available for every analysis.

Extensions and sensitivity. The same-provider Luna extension contains 150 formal runs on 75 existing retail tasks, two prompts and one repeat. Standard-prompt write-only dispatch BA is 87.5/66.6%, versus 50.0/57.0% for get/find/list reads; rewards are 32/75 and 33/75, with seven protocol-error episodes retained. This tests the operation diagnostic under another model alias.

A customer-composition check on fixed cross-fit predictions changes standard-prompt joint-constant BA from 60.5/55.5% to 59.0/53.2% under equal-customer-within-class weighting. Rules-arm return changes from 59.4% to 60.4%. Across 192 leave-one-positive-customer checks, standard-prompt cancellation ranges 57.3–62.0%; these are influence ranges, not confidence intervals.

The same-operation attribute check fixes single-item returns and a pre-fit digital-device grouping. Per prompt, 20 customer–order–item instances from 15 customers use the original customer folds and fixed LR/RF/GB learners. Maximum OOF BA is 30.3% [16.7,52.7], versus 50% for a constant predictor; a single fixed shuffled target gives 65.2%. Both prompts have identical features. Below-chance predictions can reflect reversed associations, and one shuffle is not a permutation distribution. This diagnostic is inconclusive for non-action attributes; its full predictions remain in the supplement.

Appendix D. Family-Level Holdout Evidence

Table 19 covers all 82 indexed families and 492 axis cells. T: task/input/user; M: model; C: harness/topology; H: hardware; N: network/load; D: time. H means an explicitly reported held-out unit/domain; V a condition comparison or recalibration; X a scoped restriction; ? unresolved extraction; – inapplicable to conceptual families. H identifies separation, not guaranteed transfer success. The 73 hardware and 71 temporal ? cells are not evidence of absent experiments. Core/adjacent scope is C/A. Supplement S28 retains each source locator and interpretation; the archive preserves the underlying extraction.

TABLE 17. COMPLETE INDEX OF THE 39 RESULT RECORDS (26 FAMILIES). MODE IS THE PRIMARY MEASUREMENT MODE: EX EXECUTED COLLECTION/PROTOTYPE (20), TR TRACE TRANSFORMATION (6), SI SIMULATION/DERIVED ESTIMATE (5), CO COMPONENT/CALIBRATION MEASUREMENT (6), UN PLACEMENT UNRESOLVED (2). ‡: CONTEXTUAL RECORD (6); †: ADDED BY THE FINAL SEARCH (6). T=TABLE, F=FIGURE, APP.=APPENDIX; LOCATORS REFER TO THE CITED SOURCE.

Record ID / family	Mode	Boundary; source locator
UI-delay-offline [5]	Tr	Offline delay; no live task outcome. 6.1 F5; A.5
WRA-decoy-execution [6]	Ex	Executed decoys; usefulness proxy only. 8.1–8.3 T8–11
NET-upstream-local [38]	Ex	Protected deployment/local serving; topology scoped. IV-C–E TIII–IV
NET-dummy-padding-simulation [38]	Tr	Synthetic insert/pad; no live execution. VII TIX
EARLY-K-calibration-oracle [3]	Si	K-specific fit then oracle; not live KV trial. V-A TVI; IV-A
EARLY-semantic-deidentification [3]	Ex	GPTCache prototype cost; not KV-granularity cost. V-B
WHISPER-randompadding-capture [41]	Ex	New padded captures; rounded medians. 6.1 T3–4
WHISPER-batching-replay [41]	Tr	Offline batching; model-dependent result. 6.2 F7
WHISPER-injection-replay [41]	Tr	Synthetic packets; no live zero-delay claim. 6.3 T5
SPLICE-CTBF-signal [48]	Ex	Prototype background load; tested signal only. VII-C TII
SPLICE-QCP-cost [48]	Co	QCP component cost; excludes combined CTBF. VII-C TII
TIME-length-instruction [49]	Ex	Length instruction; unequal outputs remain. 7.2
TIME-padding-estimate [49]	Si	Analytic plaintext cost; not network bytes. 7.3
SPEC-padding-REST [40]	Un	Padding/aggregation; payload cost, placement unverified. 6.1 F6–7 T2
DANGOL-capture [16]	Tr	Capture transform; distinct from live population. 9.8–9.9 T6
DANGOL-livebinding [16]	Ex	Live loopback binding; do not join capture accuracy. 9.7, OHTTP paragraph after T5
SIG-common-guardrail‡ [57]	Ex	Common instruction; not a causal guardrail effect. 4.1–4.2; App. C
SAFE-post-trigger [17]	Ex	Post-trigger probes; detector cost not service TTFT. 7.2.4 F12; 7.3–7.4
AHB-online-executed [14]	Ex	Executed UI edits; task/state changes possible. 5.1.2 T1; E.1–E.3
AHB-offline [14]	Tr	Offline touch edits; no online task damage claim. T1 offline rows; E.1–E.3
KVGOV-partialflush-sim [35]	Si	Deterministic simulation; not live protected recovery. 5.1–5.3
KVGOV-hardware-gap [35]	Co	Hardware gap calibration; not protected ASR. 5.7–5.8
KVGOV-HMAC-cost [35]	Co	HMAC microcost; not cache reuse/task loss. 5.9
TRAFFIC-prompt-reformulation [53]	Ex	Collected prompt rewrites; task utility unmeasured. V-C1 F9a
TRAFFIC-temperature [53]	Ex	Collected decoding changes; no monotone benefit. V-C1 F9a
CARLINI-fixed-rate [37]	Un	Rate/replayed cost; per-token, not request delay. VII F11–12
GAO-zero-copy-off [22]	Co	Loader component costs; not end-to-end TTFT. 7.1
TDXRAY-oblivious-map [29]	Co	Lookup microbenchmark; not full inference cost. 6.1 F8 T3
AGENTTELL-fixed-instruction‡ [64]	Ex	Live local browser tasks; loaded/cold context contrast. 4.3.1; 5 Finding 1
FP-CLOUDFLARE‡ [65]	Ex	Live access control; not user confidentiality. 6.4 T5
BAR-co-running-noise‡ [23]	Ex	MIG noise test; not an LLM defense trial. 7.2–7.3 T8
RHYTHM-VPN‡ [36]	Ex	Live VPN shift; baseline-trained attacker only. VI-B; VII
DEB-filter-origin‡ [44]	Ex	Filter creates attack view; not mitigation of it. Threat model; 4.2
CACHEPRUNE-selective† [50]	Ex	Selective cache sharing; direct token recovery/quality. 3.1; 5.1–5.6; T1; App. A
CACHEPRUNE-context† [50]	Ex	Contextual guessing; distinct from direct recovery. 3.1; 5.1–5.6; T1; App. A
OTRO-unpadded† [18]	Co	ORAM TTFT/memory; access-count padding excluded. VII-C–E; TIII
OTRO-epochpad† [18]	Si	Epoch-padding estimate; not measured padded TTFT. VII-C–E; TIII
SPARLEAK-replacement† [27]	Ex	Sparse replacement/quality; attack cost is separate. 3.1; 4–7; T4
SHAPE-padding-estimate† [28]	Si	Constant-work proposal/estimate; no executed pair. IX-B1

TABLE 18. FOUR MAIN-TEXT EXPERIMENTS AND THEIR ARTIFACT KEYS. THE RIGHT COLUMN GIVES THE SUPPLEMENT SECTION WITH IMPLEMENTATION AND FULL-GRID DETAIL.

Experiment	Artifact keys	Observation/execution boundary and comparison	Detailed source
Exp. 1: transport emission	E1, E3, E5	Buffered two-link re-emission and token-stream emitters; upstream framing, final-link framing and matched attacker refitting are distinct comparisons.	S9/S11; S12
Exp. 2: execution and link	E2, E6, E7	Prescribed stateful plans and randomized dispatch cover, then agent–tool TLS composition; dispatch-visible calls and encrypted-link records are separate views.	S9/S11; S12
Exp. 3: real trajectories and localization	E9, E10, E12	Saved autonomous Sol trajectories; original pre-test calibration, exploratory customer cross-fitting and oracle operation-category localization retain their different fit/test boundaries.	S14; S15/S21
Exp. 4: time and WF	E13, E14	Local live tool-loop timing/full tunnels, then stronger learners and specified WF adaptations; measured clocks and simulations are distinguished, with matched suites for joint comparisons.	S19; S20
Extensions	E8	Stand-in-planner pipeline check.	S13
	E11	Retrospective training-fold operating-point selection with the historical suite.	S15
	E15	Additional same-provider Luna cohort.	S22
	E16	Same-operation semantic-attribute diagnostic on reused Sol data.	S23.6
	R11/R12	Two served model endpoints: native trajectories, frozen training calibration and TLS write localization.	S24
	R14	Attacker/calibration transfer matrices, non-oracle write detection and write-only framing with the endpoint suite.	S25
	R15	Fresh paired live-planner episodes under isolated prefix cover and injected tool-response deadlines.	S26

TABLE 19. COMPLETE HOLDOUT-CODE MATRIX; CODES AND AXES ARE DEFINED IN APPENDIX D. SOURCE-QUALIFIED ROW NOTES ARE IN SUPPLEMENT S28.

Family	Scope	T	M	C	H	N	D	Family	Scope	T	M	C	H	N	D
F01 AgentPrint [4]	C	?	?	?	?	?	?	F42 ML privacy channels [44]	C	?	V	?	?	?	?
F02 UI Traces [5]	C	V	V	V	?	?	?	F43 MoE stealing [51]	C	?	?	?	?	?	?
F03 WRA [6]	C	H	?	?	?	?	?	F44 Energon [20]	C	?	V	?	?	?	?
F04 FP-Agent [65]	C	H	?	X	?	?	?	F45 SafeKV [17]	C	?	V	?	?	?	?
F05 Prompt length [1]	C	H	?	V	?	V	V	F46 InferNet [79]	A	?	V	?	V	?	?
F06 PromptPeek [2]	C	V	?	?	?	V	?	F47 ArchSnoop [26]	C	?	H	?	V	?	?
F07 NetEcho [38]	C	V	?	V	?	?	?	F48 Coding fingerprints [9]	A	X	?	X	?	?	X
F08 Early Bird [3]	C	?	V	?	?	?	?	F49 AHB [14]	C	?	?	X	?	?	?
F09 InputSnatch [46]	C	H	?	?	?	?	?	F50 WildAuth [10]	A	H	?	X	?	?	?
F10 Whisper Leak [41]	C	H	V	?	?	?	?	F51 Cycles [56]	C	X	?	X	?	?	X
F11 Spill the Beans [25]	C	V	?	?	?	?	?	F52 LumiMAS [55]	C	?	V	V	?	?	?
F12 SpliceLeak [48]	C	V	?	?	?	?	?	F53 SymDiag [80]	A	V	V	?	?	?	?
F13 Time Will Tell [49]	C	?	?	?	?	V	V	F54 LeakyLMs [47]	C	?	H	?	V	?	?
F14 Prompt Echo [39]	C	H	V	?	?	?	?	F55 KVGov [35]	C	?	?	?	V	?	?
F15 I Know What You Said [22]	C	H	V	V	V	?	?	F56 Data-centric survey [81]	A	-	-	-	-	-	-
F16 Speculation [40]	C	H	?	?	?	V	?	F57 Serving RCA [33]	C	?	?	X	?	?	?
F17 Behind Bars [23]	C	?	V	?	V	?	?	F58 ZeroPrint [11]	A	?	V	?	?	?	?
F18 Prompt caching [45]	C	?	V	?	?	?	V	F59 Endpoint fingerprints [12]	A	?	?	V	?	?	V
F19 Whose Agent [63]	C	?	?	?	?	?	?	F60 StepFinder [13]	A	H	?	?	?	?	?
F20 Workflow traffic [16]	C	?	?	H	?	?	?	F61 Web-agent detection [54]	C	?	V	?	?	?	?
F21 GPU contention [42]	C	?	?	V	?	V	?	F62 Cross-campaign [61]	C	V	?	V	?	?	?
F22 AgentTell [64]	C	V	?	?	?	V	?	F63 AgentProv [59]	C	?	V	?	?	?	?
F23 Traffic benchmark [53]	C	?	?	H	?	?	?	F64 LiDAR [60]	C	?	V	V	?	?	?
F24 Remote timing [37]	C	?	V	?	?	?	V	F65 LLM-WFIN [32]	C	V	?	?	?	?	?
F25 Persona fingerprint [62]	C	?	?	X	?	?	?	F66 MIRAGE [52]	C	V	V	?	?	?	?
F26 TDXRay [29]	C	?	?	?	?	?	?	F67 Routing metadata [43]	C	V	?	V	?	?	?
F27 Detokenization [31]	C	?	?	X	?	?	?	F68 Shape and Substance [28]	C	V	V	V	V	?	?
F28 Spillage [82]	A	V	?	?	?	?	?	F69 SplitDecoy [34]	C	?	?	?	?	?	?
F29 PlanTwin [83]	A	V	?	?	?	?	?	F70 PAGE404 [24]	C	V	V	?	?	?	?
F30 SICC [84]	A	?	?	?	?	?	?	F71 Edge EM [19]	C	H	?	?	?	?	?
F31 TOPR [85]	A	V	?	?	?	?	?	F72 SparLeak [27]	C	V	V	V	?	?	?
F32 ClawGuard [15]	C	?	?	V	?	?	V	F73 Sparseety [30]	C	V	V	V	?	?	?
F33 SIGLEAK [57]	C	V	?	?	?	?	?	F74 OTRO [18]	C	?	?	?	?	?	?
F34 ProcGrep [58]	C	H	?	V	?	?	?	F75 CachePrune [50]	C	V	V	?	?	?	?
F35 SecEmb [86]	A	?	X	?	?	?	?	F76 Privacy foundations [68]	A	-	-	-	-	-	-
F36 Mobile agents [87]	A	?	?	?	?	?	?	F77 DP-MemView [88]	A	?	?	?	?	?	?
F37 Hidden risks [89]	A	?	?	?	?	?	?	F78 PriCon [90]	A	?	?	?	?	?	?
F38 Browser privacy [91]	A	?	?	V	?	?	?	F79 SCAAR [92]	A	?	?	?	?	?	?
F39 LLMs Have Rhythm [36]	C	?	V	?	?	H	V	F80 CacheSolidarity [93]	A	?	?	?	?	?	?
F40 MoEcho [21]	C	H	V	?	?	V	?	F81 Vision encoder [94]	A	?	?	?	?	?	?
F41 LLM-Fin [72]	A	?	H	?	?	?	?	F82 Split gradients [95]	A	?	?	?	?	?	?